

Calibration Science

Science

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Short Title:	Calibration Science	
Faculty	Science and Computing	
Department	Science	
Cluster	Quality, Regulatory and Compliance	
Author	ADONEGAN	
Credits	5	Level: Introductory

Description of Module / Aims

This module covers all aspects of calibration within the pharmaceutical industry. Students will learn and observe the practice of performing calibration on routine items of equipment. They will be given the skills to develop a calibration schedule taking into consideration the frequency, how critical the equipment is to the manufacturing process and is GMP compliant.

Programmes

CALI-0001 Higher Certificate in Science in Good Manufacturing Practice and Technology 2 / 4 / M
(WD_SGMPT_C)

Indicative Content

- Definition of calibration; review of the purpose of calibration and frequency of calibration; the role of the National Standards Authority of Ireland (NSAI) and international bodies
- Review of GMP requirements for equipment calibration and the maintenance of calibration documentation (procedure and records); identification of calibrated from non-calibrated equipment
- Measurement techniques mass, temperature dimension, pressure and the extent of these in pharmaceutical industry - calibration of instruments for these measurements
- Setting up logbooks for the calibration
- Choose from the following list of equipment: analytical balance, callipers, pH meter, conductivity meter, hygrometers, UV-visible spectrometers, microscopes
- Volumetric flasks and graduated cylinders, pipettes, micro-pipettes, thermometers, autoclaves, incubators, fridges, freezers
- Calibration, operation and maintenance of equipment
- Primary and secondary calibration in temperature measurement

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Discuss the principles of calibration in Quality Assurance and the external agencies involved.
2. Distinguish between calibrated and non-calibrated equipment, calibration certificates and records.
3. Explain the principles of measurement and techniques involved for routine process use and for calibration.
4. Summarise a calibration procedure and carry out a calibration of a simple item of equipment.

Learning and Teaching Methods

- Lectures.
- Discussion.
- Practical examples.
- Industrial visits.

- E-learning via Moodle.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4
<i>Lab Report</i>	25%	4
<i>In-Class Assessment</i>	75%	1,2,3

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		24
Independent Learning		111

Supplementary Material

- Morris, A. *Measurement and calibration requirements for quality assurance to ISO9000*. Chichester: John Wiley, 1997.
- Placko, D. *Fundamentals of Instrumentation and Measurement*. Chichester: Wiley, 2008.

Requested Resources

- Room Type: Chemistry Lab